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Course Name: Data Handling and Visualization	Course Name: Data Handling and Visualization

DATA VISUALIZATION — SET 16 TO 20

SET 16: CUSTOMER DEMOGRAPHICS ANALYSIS

Dataset: *Customer Demographics*

Customer ID	Age	Gender	Income (in \$)
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

Q1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.

CODE:

```
library(ggplot2)
customer_data <- data.frame(
  CustomerID = c(1, 2, 3),
  Age = c(28, 35, 42)
)
ggplot(customer_data, aes(x = factor(Age), y = Age, fill = factor(Age))) +
  geom_bar(stat = "identity") +
  labs(title = "Distribution of Customer Ages",
       x = "Customer", y = "Age") +
  theme_minimal() +
  theme(legend.position = "none")
```

OUTPUT:

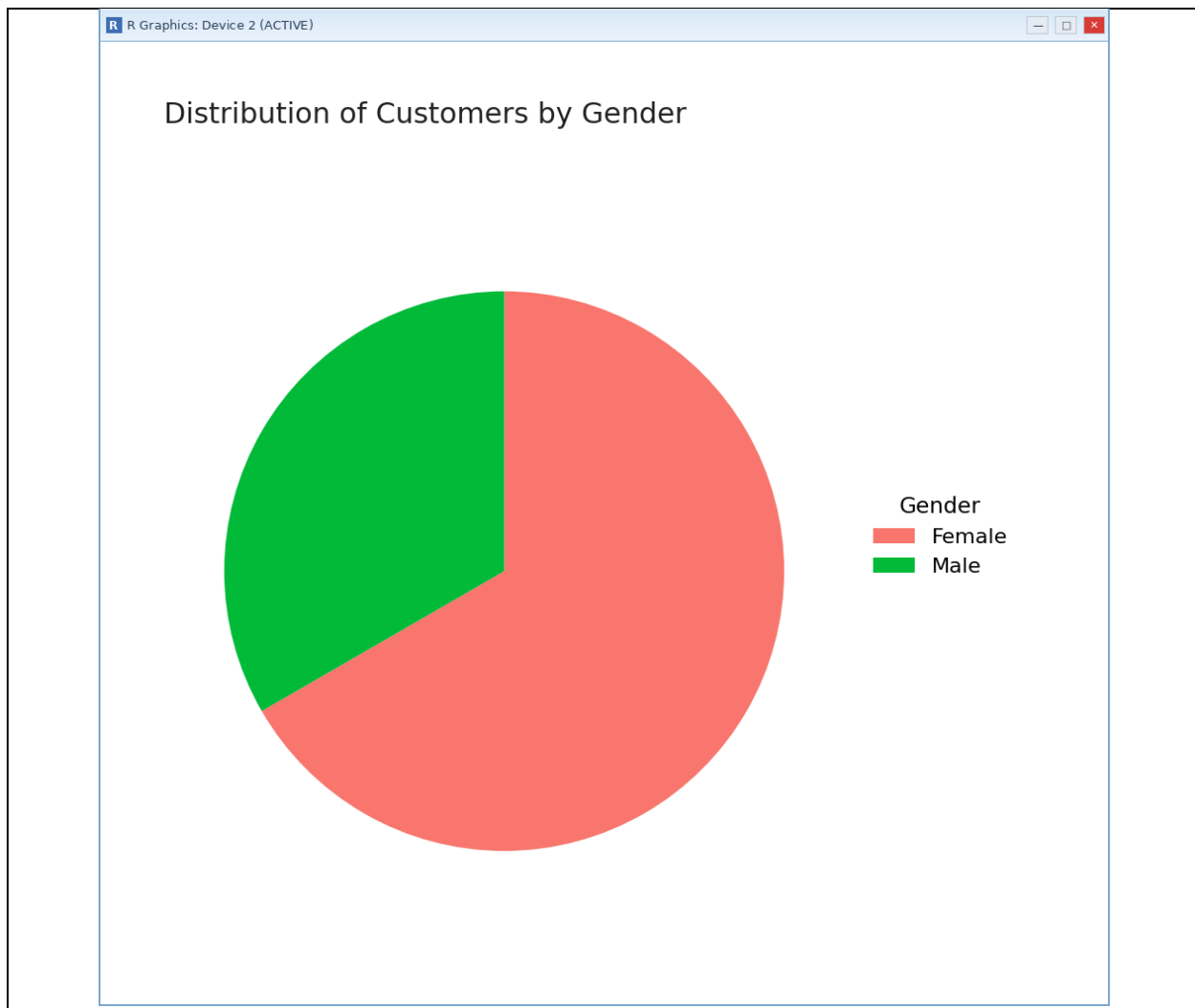


Q2. Generate a pie chart to represent the distribution of customers by gender.

CODE:

```
library(ggplot2)
gender_data <- data.frame(
  Gender = c("Female", "Male"),
  Count = c(2, 1)
)
ggplot(gender_data, aes(x = "", y = Count, fill = Gender)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y", start = 0) +
  labs(title = "Distribution of Customers by Gender") +
  theme_void()
```

OUTPUT:



Q3. Build a table to show the demographic information for each customer. Label the table elements.

CODE:

```
customer_data <- data.frame(  
  Customer_ID = c(1, 2, 3),  
  Age = c(28, 35, 42),  
  Gender = c("Female", "Male", "Female"),  
  Income = c(50000, 60000, 75000)  
)  
print(customer_data)
```

OUTPUT:

	Customer ID	Age	Gender	Income
	1	28	Female	50000.00
	2	35	Male	60000.00
	3	42	Female	75000.00

SET 17: EMPLOYEE PERFORMANCE ANALYSIS

Dataset: *Employee Performance*

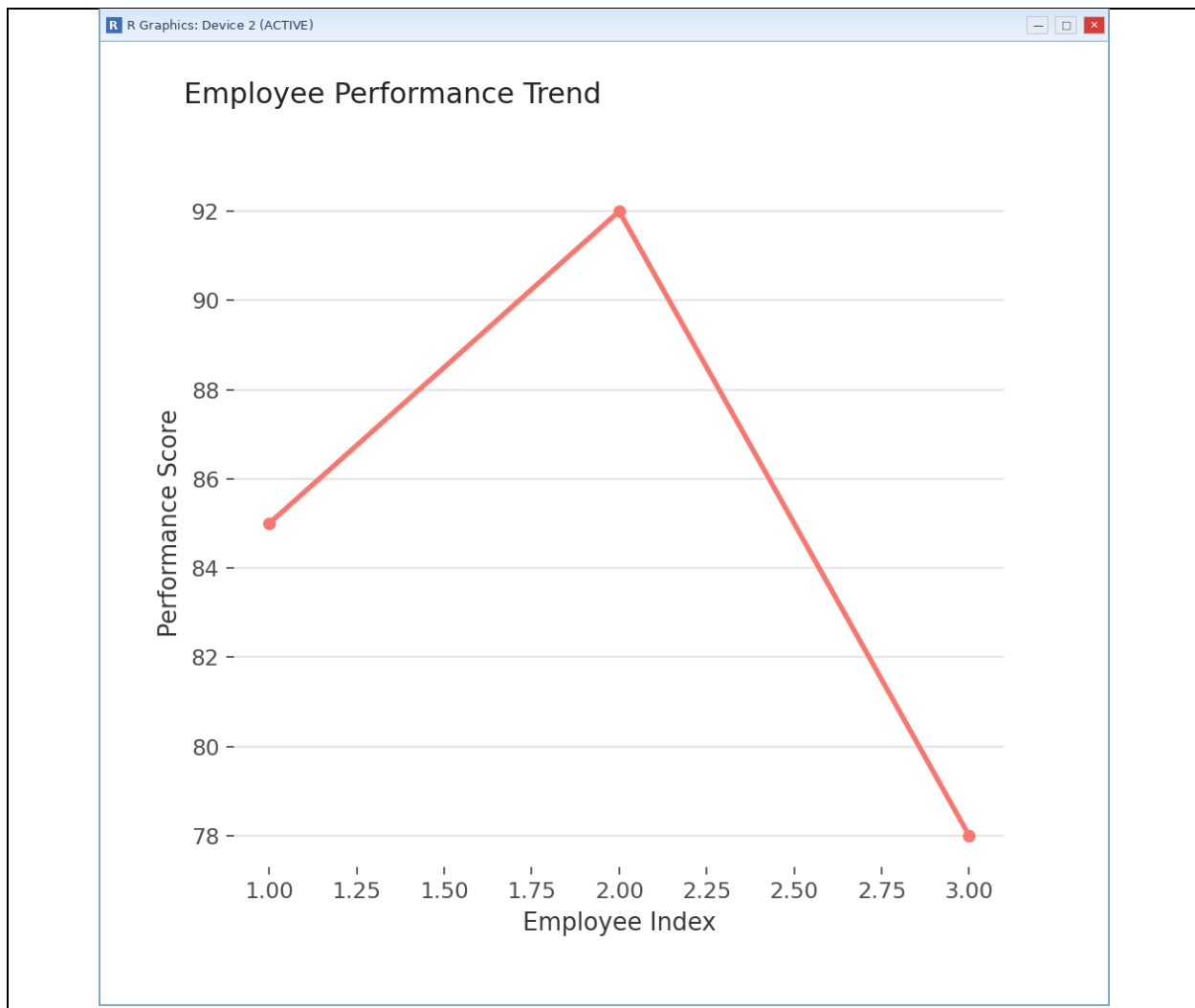
Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

Q1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

CODE:

```
library(ggplot2)
employee_data <- data.frame(
  EmployeeID = c(1, 2, 3),
  Performance = c(85, 92, 78)
)
ggplot(employee_data, aes(x = EmployeeID, y = Performance)) +
  geom_line(color = "#F8766D", linewidth = 1.2) +
  geom_point(color = "#F8766D", size = 2) +
  labs(title = "Employee Performance Trend",
       x = "Employee Index", y = "Performance Score") +
  theme_minimal()
```

OUTPUT:

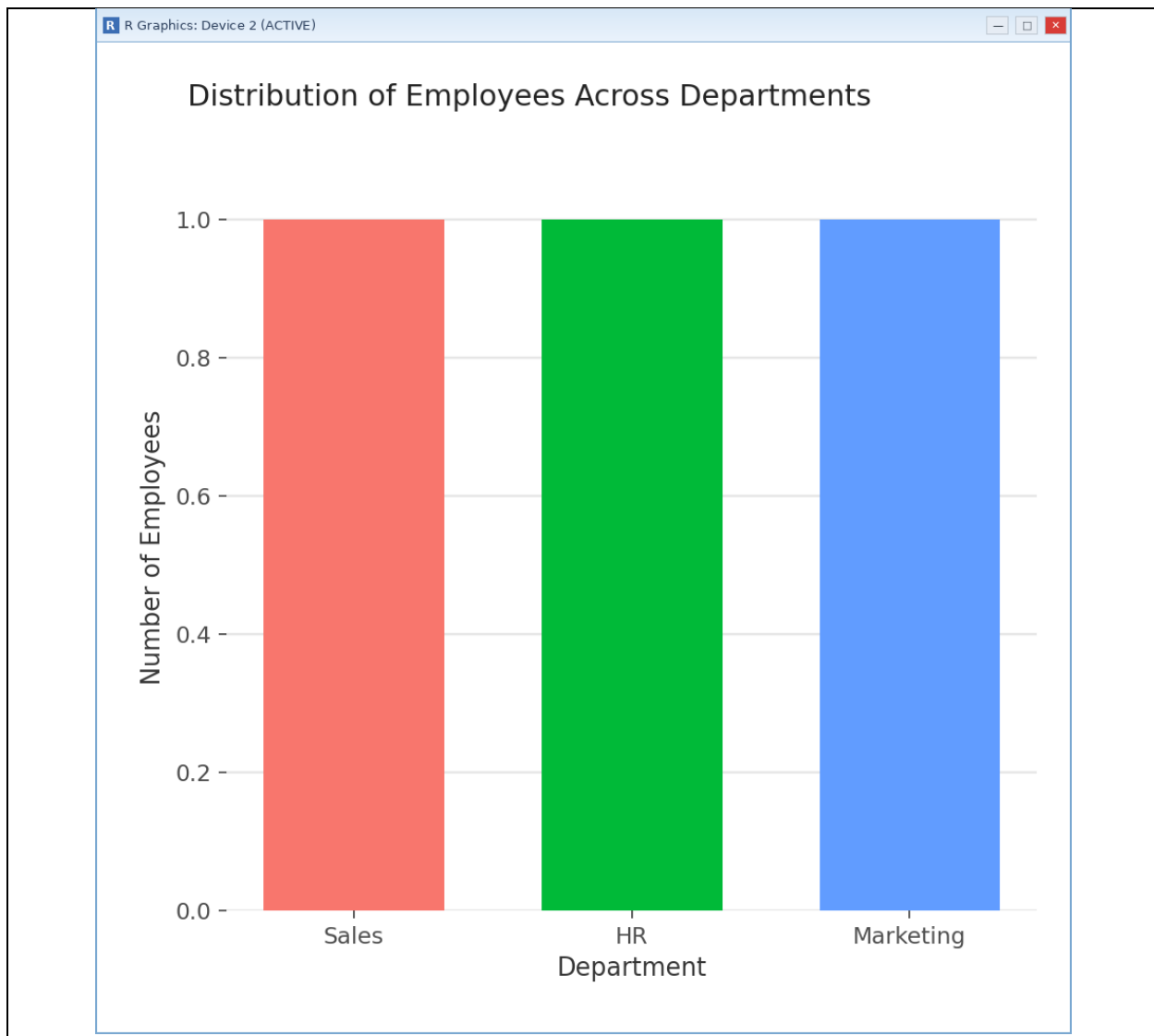


Q2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

CODE:

```
library(ggplot2)
dept_data <- data.frame(
  Department = c("Sales", "HR", "Marketing"),
  Count = c(1, 1, 1)
)
ggplot(dept_data, aes(x = Department, y = Count, fill = Department)) +
  geom_bar(stat = "identity") +
  labs(title = "Distribution of Employees Across Departments",
       x = "Department", y = "Number of Employees") +
  theme_minimal() +
  theme(legend.position = "none")
```

OUTPUT:



Q3. Build a table to display the performance data for each employee. Label the table elements.

CODE:

```
employee_data <- data.frame(  
  Employee_ID = c(1, 2, 3),  
  Department = c("Sales", "HR", "Marketing"),  
  Years = c(5, 3, 7),  
  Score = c(85, 92, 78)  
)  
print(employee_data)
```

OUTPUT:

	Employee ID	Department	Years	Score
	1	Sales	5	85
	2	HR	3	92
	3	Marketing	7	78

SET 18: PRODUCT INVENTORY MANAGEMENT

Dataset: *Product Inventory*

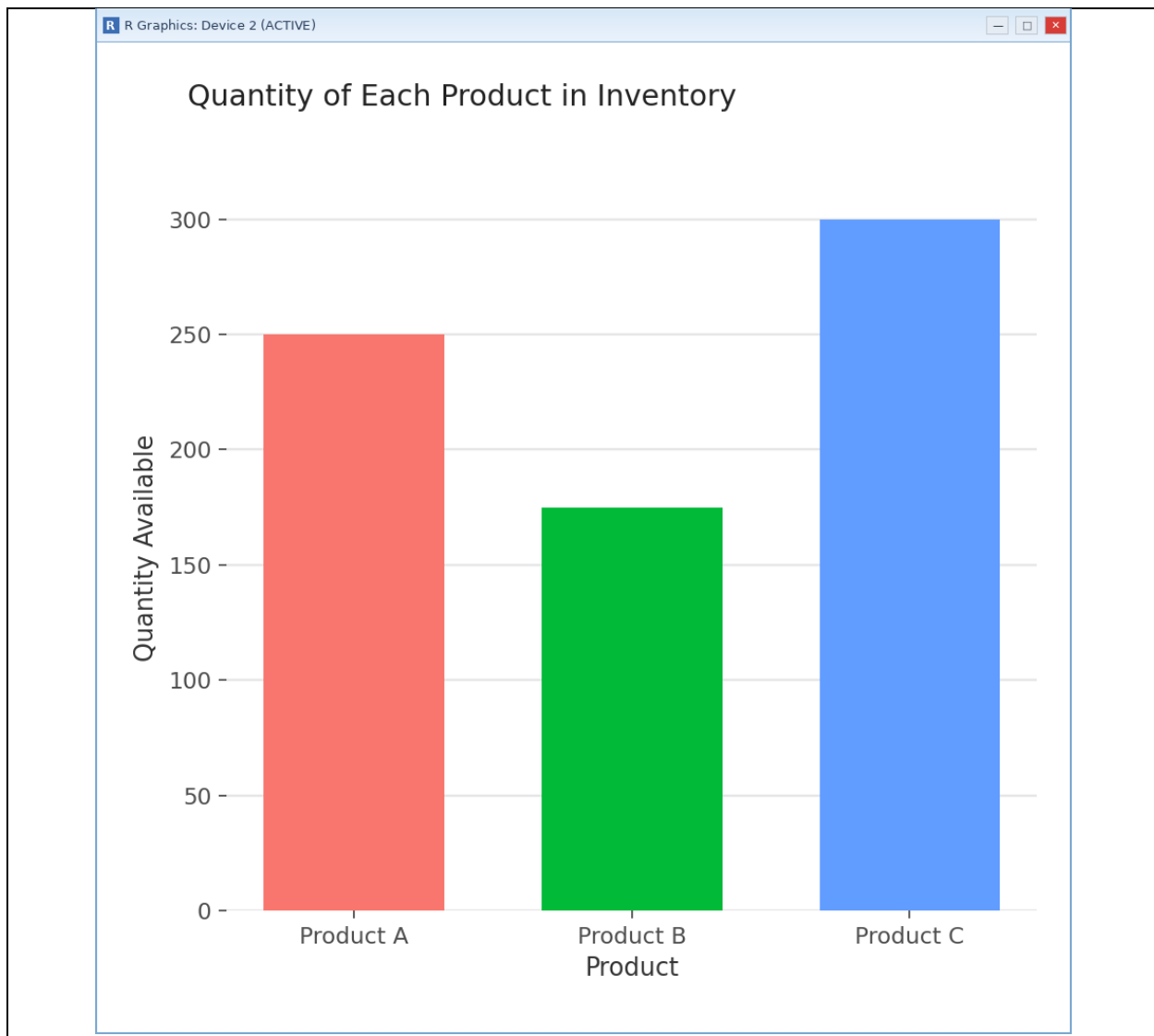
Product ID	Product Name	Quantity Available	Price (in \$)
1	Product A	250	20
2	Product B	175	15
3	Product C	300	18

Q1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

CODE:

```
library(ggplot2)
inventory_data <- data.frame(
  Product = c("Product A", "Product B", "Product C"),
  Quantity = c(250, 175, 300)
)
ggplot(inventory_data, aes(x = Product, y = Quantity, fill = Product)) +
  geom_bar(stat = "identity") +
  labs(title = "Quantity of Each Product in Inventory",
       x = "Product", y = "Quantity Available") +
  theme_minimal() +
  theme(legend.position = "none")
```

OUTPUT:



Q2. Generate a stacked bar chart to show the quantity of each product within different product categories.

CODE:

```
library(ggplot2)
inventory_data <- data.frame(
  Category = c("Category 1", "Category 1", "Category 2"),
  Product = c("Product A", "Product C", "Product B"),
  Quantity = c(250, 300, 175)
)
ggplot(inventory_data, aes(x = Category, y = Quantity, fill = Product)) +
  geom_bar(stat = "identity") +
  labs(title = "Product Quantity by Category",
       x = "Product Category", y = "Quantity Available") +
  theme_minimal()
```


OUTPUT:



Q3. Build a table to show the inventory data for each product. Label the table elements.

CODE:

```
inventory_data <- data.frame(  
  Product_ID = c(1, 2, 3),  
  Name = c("Product A", "Product B", "Product C"),  
  Qty = c(250, 175, 300),  
  Price = c(20, 15, 18)  
)  
print(inventory_data)
```

OUTPUT:

	Product ID	Name	Qty	Price
	1	Product A	250	20.00
	2	Product B	175	15.00
	3	Product C	300	18.00

SET 19: SURVEY RESPONSES ANALYSIS

Dataset: *Survey Responses*

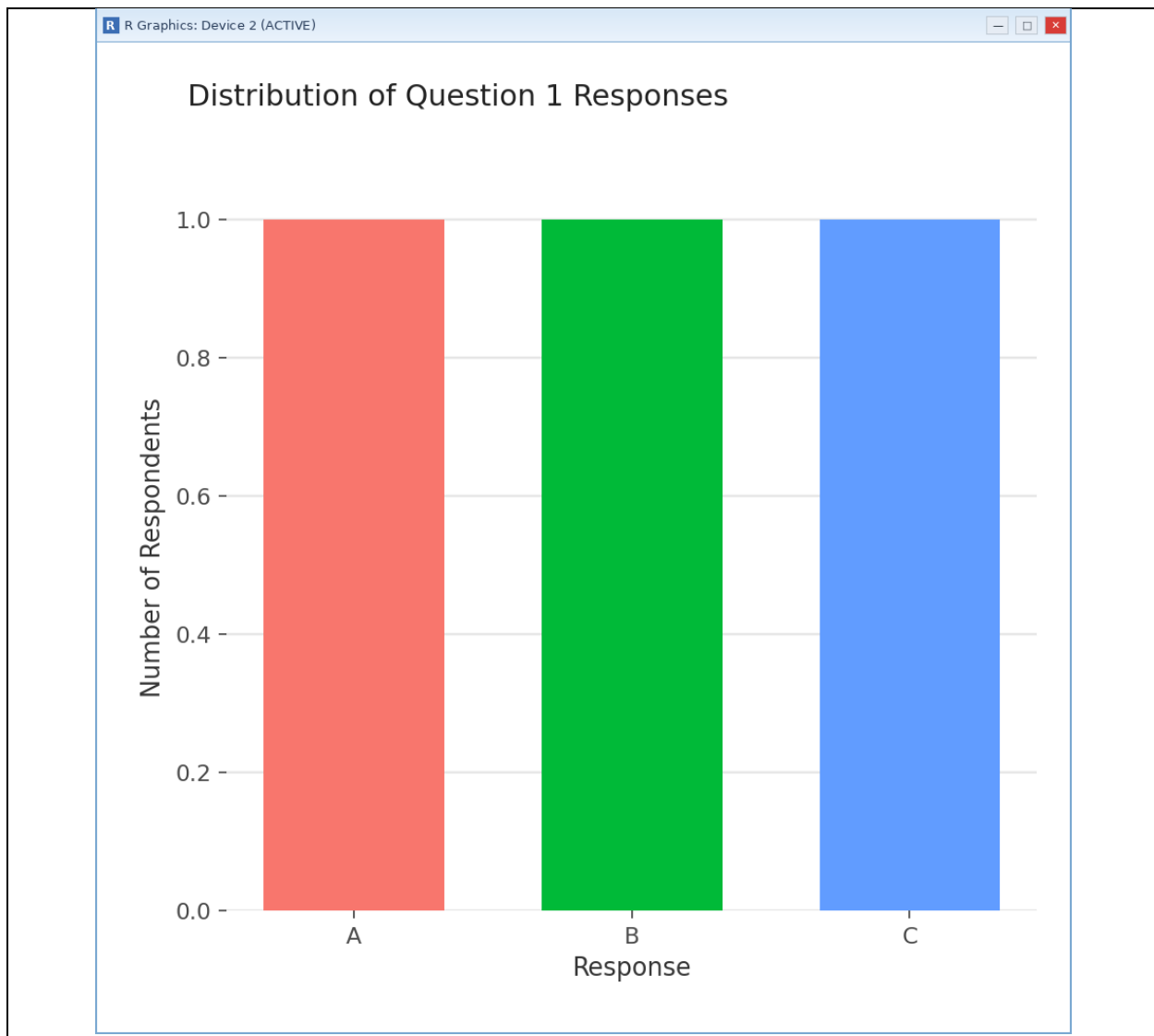
Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

Q1. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.

CODE:

```
library(ggplot2)
q1_data <- data.frame(
  Response = c("A", "B", "C"),
  Count = c(1, 1, 1)
)
ggplot(q1_data, aes(x = Response, y = Count, fill = Response)) +
  geom_bar(stat = "identity") +
  labs(title = "Distribution of Question 1 Responses",
       x = "Response", y = "Number of Respondents") +
  theme_minimal()
```

OUTPUT:

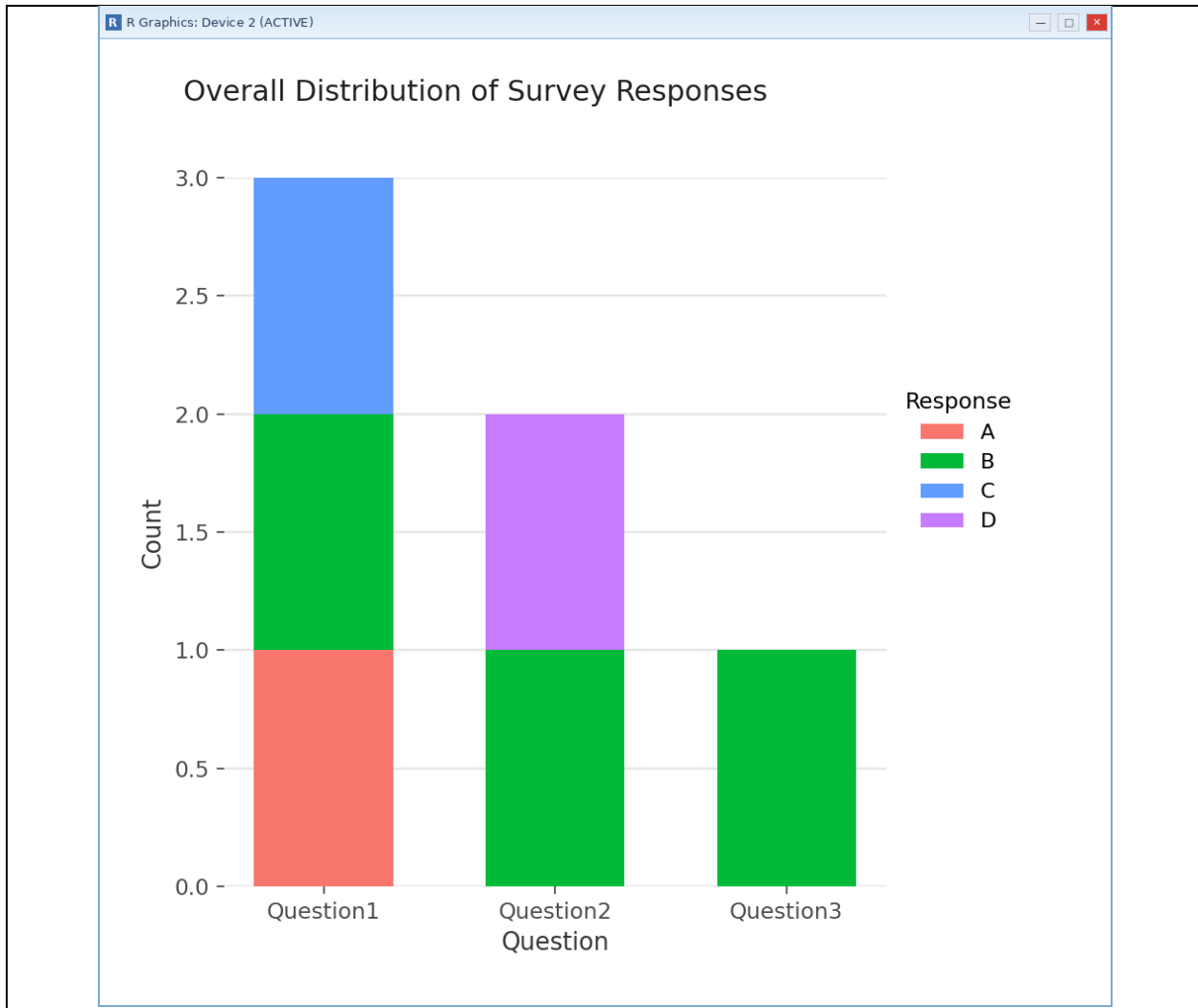


Q2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

CODE:

```
library(ggplot2)
survey_long <- data.frame(
  Question = c("Question1","Question1","Question1",
               "Question2","Question2","Question2",
               "Question3","Question3","Question3"),
  Response = c("A","B","C","B","A","D","C","B","A")
)
ggplot(survey_long, aes(x = Question, fill = Response)) +
  geom_bar() +
  labs(title = "Overall Distribution of Survey Responses",
       x = "Question", y = "Count") +
  theme_minimal()
```

OUTPUT:



Q3. Build a table to show the survey response data for each survey. Label the table elements.

CODE:

```
survey_data <- data.frame(  
  Survey_ID = c(1, 2, 3),  
  Q1 = c("A", "B", "C"),  
  Q2 = c("B", "A", "A"),  
  Q3 = c("C", "D", "B")  
)  
print(survey_data)
```

OUTPUT:

	Survey ID	Q1	Q2	Q3
	1	A	B	C
	2	B	A	D
	3	C	A	B

SET 20: STOCK ANALYSIS

Dataset: *Stock Prices*

Date	Stock A	Stock B	Stock C
2023-01-01	100	150	120
2023-01-02	105	152	118
2023-01-03	110	148	122

Q1. Create a line chart to visualize the stock prices of three companies (Stock A, Stock B, and Stock C) over a specific time period. Label the axes and title the chart.

CODE:

```
library(ggplot2)
library(tidyr)
stock_data <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
  StockA = c(100, 105, 110),
  StockB = c(150, 152, 148),
  StockC = c(120, 118, 122)
)
stock_long <- pivot_longer(stock_data, cols = -Date,
                           names_to = "Stock", values_to = "Price")
ggplot(stock_long, aes(x = Date, y = Price, color = Stock)) +
  geom_line(linewidth = 1.2) + geom_point(size = 2) +
  labs(title = "Stock Prices Over Time",
       x = "Date", y = "Stock Price") +
  theme_minimal()
```

OUTPUT:

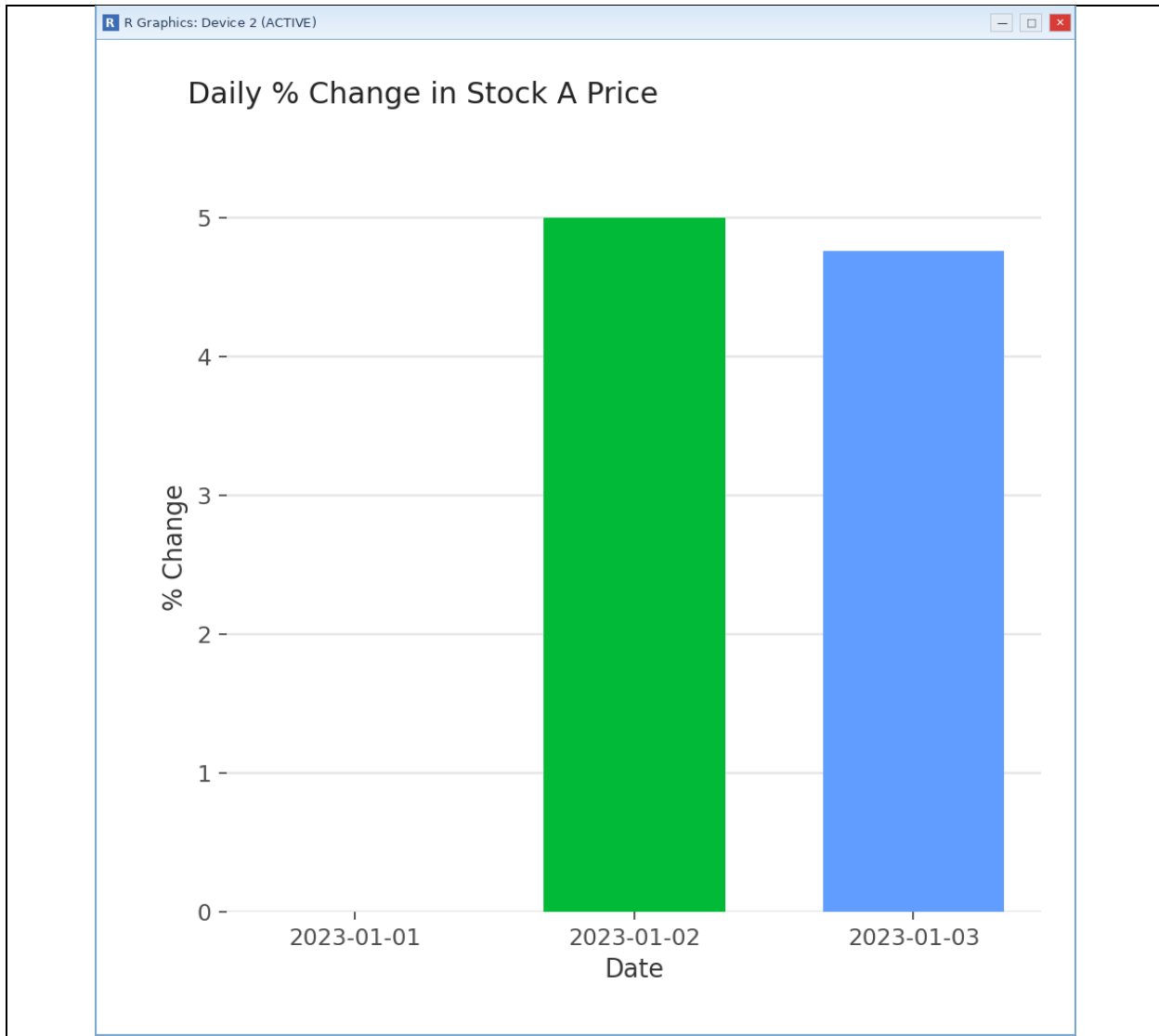


Q2. Generate a bar chart showing the daily percentage change in stock prices for Stock A. Label the chart elements.

CODE:

```
library(ggplot2)
stock_a <- c(100, 105, 110)
pct_change <- c(0, diff(stock_a) / head(stock_a, -1) * 100)
change_data <- data.frame(
  Date = c("2023-01-01", "2023-01-02", "2023-01-03"),
  PctChange = round(pct_change, 2)
)
ggplot(change_data, aes(x = Date, y = PctChange, fill = Date)) +
  geom_bar(stat = "identity") +
  labs(title = "Daily % Change in Stock A Price",
       x = "Date", y = "% Change") +
  theme_minimal() +
  theme(legend.position = "none")
```

OUTPUT:



Q3. Build a table to display the stock price data for each company over the given period. Label the table elements.

CODE:

```
stock_data <- data.frame(  
  Date = c("2023-01-01", "2023-01-02", "2023-01-03"),  
  Stock_A = c(100, 105, 110),  
  Stock_B = c(150, 152, 148),  
  Stock_C = c(120, 118, 122)  
)  
print(stock_data)
```

OUTPUT:

Date	Stock A	Stock B	Stock C
2023-01-01	100	150	120
2023-01-02	105	152	118
2023-01-03	110	148	122